

	$\Delta U = q - W$ compare with $\Delta U = Q + W$ (usual convention) W is work done by system
13	Ans: D For SHM, $a = -\omega^2 x$ Hence, magnitude of restoring force = ma $= m [(2\pi f)^2 x]$ $= 0.040 \times (2\pi \times 25)^2 \times 0.0025$ $= 2.47 \text{ N}$
14	Ans: C